

EKG/ECG INTERPRETATION GUIDE

Quick Reference for EMS Professionals

Normal EKG Values

Component	Normal Range	Description
Rate	60-100 bpm	Normal sinus rhythm
PR Interval	0.12-0.20 sec (3-5 small boxes)	Time from atrial to ventricular depolarization
QRS Complex	< 0.12 sec (< 3 small boxes)	Ventricular depolarization
QT Interval	< 0.44 sec (men), < 0.46 sec (women)	Ventricular depolarization + repolarization
P Wave	Upright in II, III, aVF	Atrial depolarization
QRS Axis	-30 to +90 degrees	Normal electrical axis

Systematic EKG Interpretation (8 Steps)

1. Rate

Count R waves x 10 (300/150/100/75/60/50) or 300 / large boxes between R waves

2. Rhythm

Regular? Irregular? Regularly irregular? P waves present? Every P followed by QRS?

3. P Waves

Present? Uniform? Upright in II? One P per QRS?

4. PR Interval

Normal (0.12-0.20s)? Constant? Prolonged? Shortened?

5. QRS Complex

Narrow (<0.12s) or Wide (>=0.12s)? Pathological Q waves?

6. ST Segment

Elevated? Depressed? Isoelectric?

7. T Waves

Upright? Inverted? Peaked? Flattened?

8. QT Interval

Corrected QT (QTc) < 440ms men, < 460ms women?

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Critical EKG Findings - RECOGNIZE & ACT

Finding	What to Look For	Significance	Action
STEMI	ST elevation $\geq$ 1mm in 2+ contiguous walls	Active MI	Activate cath lab, MONA, ASA
STEMI Equiv	ST depression in V1-V3 (posterior), New LBBB	Posterior MI / MI	Treat as STEMI
VTach	Wide QLS, rate >100 , AV dissociation	Unstable if symptomatic	Amiodarone, shock if unstable
VFib	Chaotic, no organized rhythm	Cardiac arrest	Defibrillate immediately
Asystole	Flat line (confirm in 2 leads)	Cardiac arrest	CPR, epi, search cause
PEA	Organized rhythm, no pulse	Cardiac arrest	CPR, epi, search reversible causes
3rd Degree HB	P and QRS dissociated, junctional escape	Unstable bradycardia	Transcutaneous pacing
AFlutter	Sawtooth pattern, atrial rate ~ 300	Unstable if rapid	Rate control, cardioversion
Torsades	Polymorphic VT, prolonged QT	Degenerates to VF	Magnesium 2g IV
HyperK	Peaked Ts, wide QRS, sine wave	Cardiac arrest risk	Calcium, insulin/D50, albuterol
Brugada	RBBB + STE V1-V3	Sudden death risk	Avoid certain drugs, ICD
WPW	Delta wave, short PR, wide QLS	Risk of rapid SVT/Afib	Avoid AV nodal blockers

12-Lead STEMI Localization

Leads Involved	Wall	Artery	Notes
V1-V2	Septal	LAD (septal)	R in V1 may indicate posterior MI
V3-V4	Anterior	LAD	Most common STEMI location
V5-V6 + I, aVL	Lateral	LCx or LAD diagonal	Check for reciprocal changes inferiorly
II, III, aVF	Inferior	RCA (90%) or LCx	Always get right leads for RV involvement
V3R-V4R	Right Ventricle	RCA (proximal)	Hypotension + clear lungs = consider RV MI
V7-V9	Posterior	RCA or LCx	Look for ST depression in V1-V3 as mirror

Normal Sinus Rhythm Criteria

- Rate: 60-100 bpm
- Rhythm: Regular
- P waves: Upright in II, III, aVF
- Every P wave followed by a QRS
- PR interval: 0.12-0.20 seconds (.constant)
- QRS: Narrow (< 0.12 seconds)
- One P wave for every QRS complex

Common Dysrhythmias Quick Reference

Rhythm	Rate	QRS	P Wave	Treatment
Sinus Brady	<60	Narrow	Normal	Atropine 0.5mg, TCP if symptomatic
Sinus Tach	100-150	Narrow	Normal	Treat underlying cause
PSVT	150-250	Narrow	Hidden	Vagal, Adenosine 6/12mg
AFlutter	Variable	Narrow	Sawtooth	Rate control, anticoagulate
AFib	Irregular	Narrow	None/fibs	Rate/rhythm control, anticoagulate
VTach	100-250	Wide	None/dissoc	Amiodarone/lidocaine, shock if unstable
Torsades	200-250	Wide, changing	None	Magnesium 2g IV, overdrive pacing
VFib	None	Chaotic	None	Defibrillate, CPR, epi
Asystole	None	None	None	CPR, epi, search causes
PEA	Variable	Variable	May have	CPR, epi, search reversible causes

DISCLAIMER: Educational purposes only. Not medical advice. Always follow local protocols and medical direction. Verify all clinical information against current evidence-based resources.